

Online Quiz System - Sprint 1

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S1-01 Database Schema

Database created and sample data loaded.

```
(venv) PS C:\Users\DELL\Desktop\quiz_system> python init_db.py
User: 2 rows
Category: 3 rows
Quiz: 1 rows
Question: 5 rows
Result: 0 rows
quiz.db created.
(venv) PS C:\Users\DELL\Desktop\quiz_system>
```

All five tables are created in the database.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r[0]) for r in sqlite3.connect('quiz.db').execute('SELECT name FROM sqlite_master') if not r[0].startswith('sqlite')]"
User
Category
Quiz
Question
Result
PS C:\Users\DELL\Desktop\quiz_system>
```

Category, Quiz and Question tables are connected.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r) for r in sqlite3.connect('quiz.db').execute('SELECT c.name, q.title, q.time_limit, COUNT(qu.question_id) FROM Category c JOIN Quiz q ON q.category_id=c.category_id JOIN Question qu ON qu.quiz_id=q.quiz_id GROUP BY q.quiz_id')]"
('Java', 'Java Basics Quiz', 10, 5)
PS C:\Users\DELL\Desktop\quiz_system>
```

Question table fields match the SRS.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r[1], r[2]) for r in sqlite3.connect('quiz.db').execute('PRAGMA table_info(Question)')]"
question_id INTEGER
quiz_id INTEGER
question_text VARCHAR(500)
option_a VARCHAR(200)
option_b VARCHAR(200)
option_c VARCHAR(200)
option_d VARCHAR(200)
correct_option CHAR(1)
PS C:\Users\DELL\Desktop\quiz_system> |
```

Passwords are stored in hashed form.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r[0], r[1][:25]+'...') for r in sqlite3.connect('quiz.db').execute('SELECT email, password FROM User')]"
testuser01@example.com scrypt:32768:8:1$ZR2WLGQM...
admin01@example.com scrypt:32768:8:1$9JHJJB9...
poojitha.test@example.com scrypt:32768:8:1$PEyo3J1b...
PS C:\Users\DELL\Desktop\quiz_system> |
```

S1-02 Registration and Login

App files unzipped and the server started.

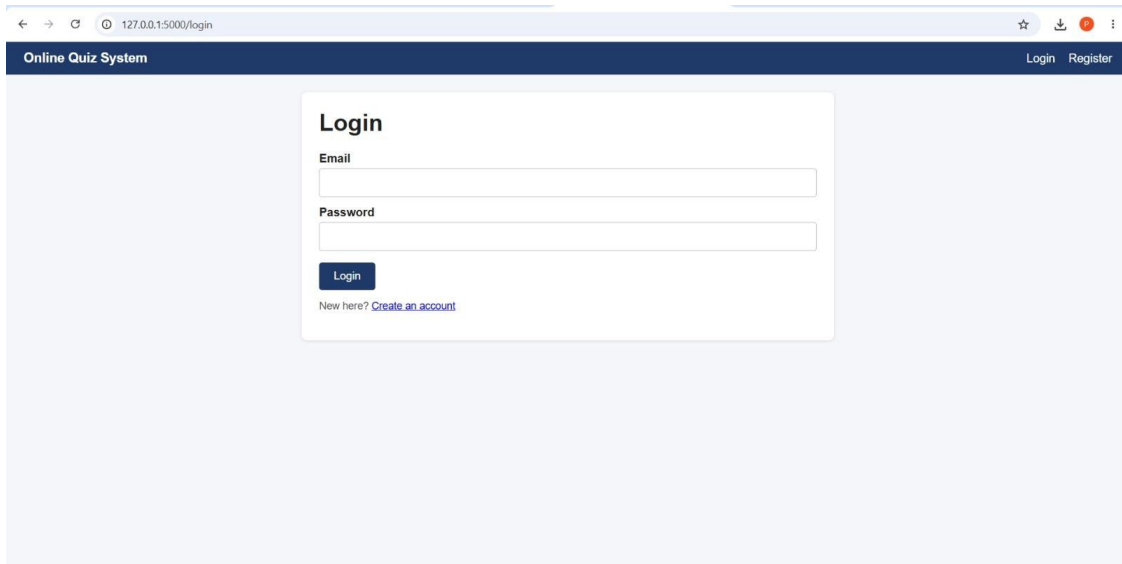
```
(venv) PS C:\Users\DELL\Desktop\quiz_system> Expand-Archive "$HOME\Downloads\S1-02_login_files.zip" -DestinationPath . -Force
(venv) PS C:\Users\DELL\Desktop\quiz_system> dir

Directory: C:\Users\DELL\Desktop\quiz_system

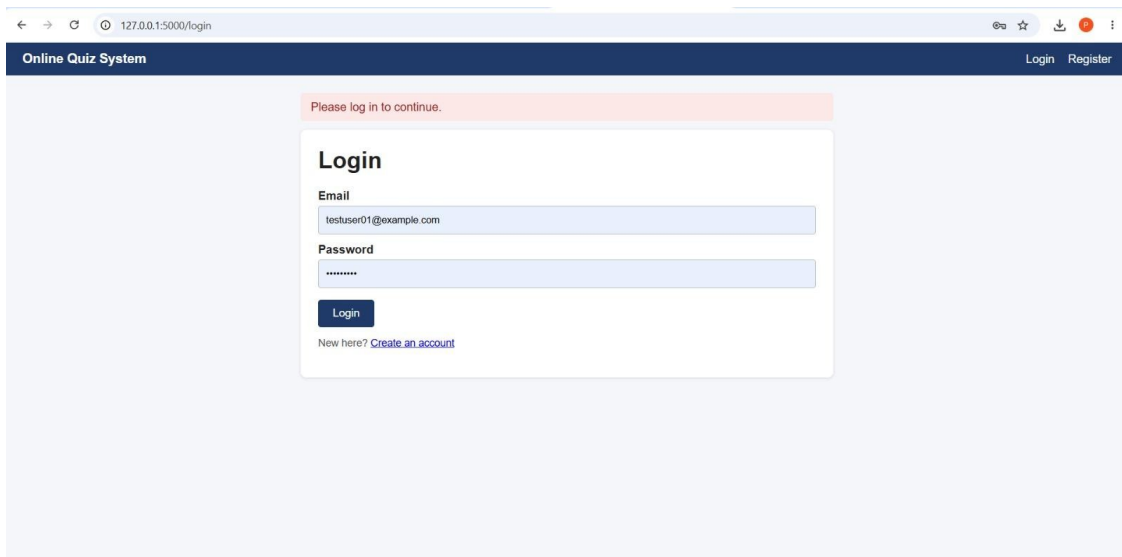
Mode                LastWriteTime         Length Name
----                -
d-----         10/9/2026   12:18 PM             static
d-----         10/9/2026   12:18 PM            templates
d-----         10/9/2026   11:25 AM             venv
-a-----         10/9/2026    6:15 AM          4163 app.py
-a-----         10/9/2026   11:11 AM          2522 init_db.py
-a-----         10/9/2026   11:40 AM          36864 quiz.db
-a-----         10/9/2026   11:10 AM          1972 schema.sql

(venv) PS C:\Users\DELL\Desktop\quiz_system> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 296-466-228
```

Login page.



TC-22: Opening a page without login goes to the login page.



Registration successful.

A screenshot of a web browser showing the 'Online Quiz System' login page. The browser's address bar displays '127.0.0.1:5000/login'. A green notification banner at the top states 'Registration successful. Please log in.' The main form is titled 'Login' and contains two input fields: 'Email' with the value 'poojitha.test@example.com' and 'Password' with masked characters '.....'. A 'Login' button is positioned below the password field. At the bottom of the form, there is a link that says 'New here? [Create an account](#)'.

Password with less than 6 characters is not accepted.

A screenshot of the 'Online Quiz System' registration page. The browser's address bar shows '127.0.0.1:5000/register'. A red error banner at the top reads 'Password must be at least 6 characters.' The 'Create an account' form includes fields for 'Name' (Poojitha), 'Email' (short.test@example.com), 'Password (min 6 characters)' (which is empty), and 'Confirm password' (also empty). A 'Register' button is at the bottom of the form, and a link below it says 'Already registered? [Login](#)'.

Passwords that do not match are not accepted.

A screenshot of the 'Online Quiz System' registration page. The browser's address bar shows '127.0.0.1:5000/register'. A red error banner at the top states 'Passwords do not match.' The 'Create an account' form shows 'Name' as 'Poojitha' and 'Email' as 'testuser01@example.com'. The 'Password (min 6 characters)' field contains masked characters '.....', while the 'Confirm password' field is empty. A 'Register' button is located at the bottom of the form, with a link below it that says 'Already registered? [Login](#)'.

An email that already exists is not accepted.

Online Quiz System Login Register

An account with this email already exists.

Create an account

Name
Poojitha

Email
poojitha.test@example.com

Password (min 6 characters)

Confirm password

[Register](#)

Already registered? [Login](#)

TC-26: Wrong password shows an error.

Online Quiz System Login Register

Invalid email or password.

Login

Email
testuser01@example.com

Password

[Login](#)

New here? [Create an account](#)

Login successful. Home page is shown.

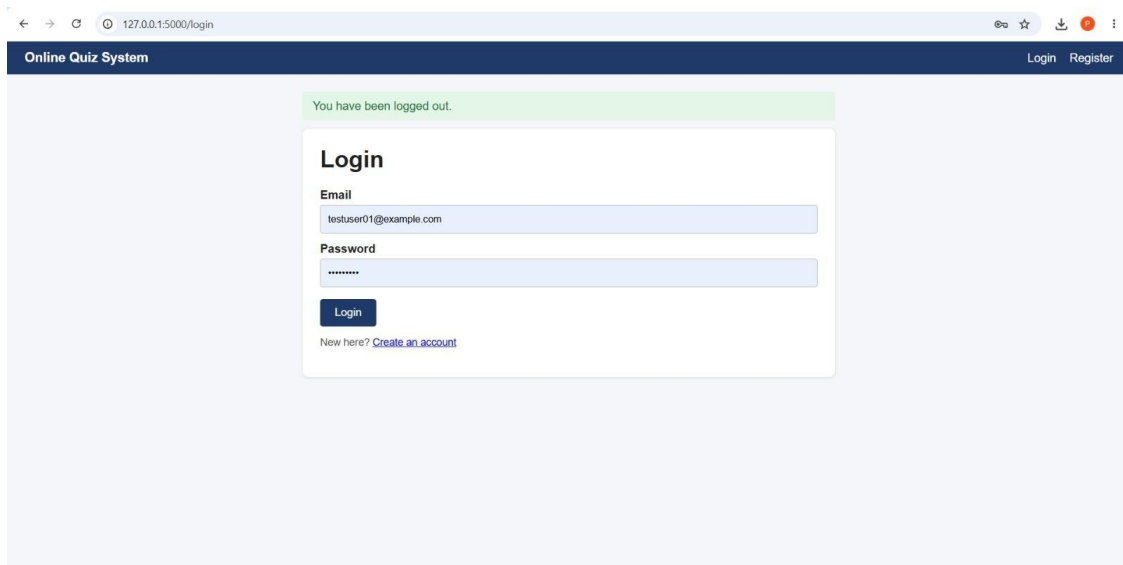
Online Quiz System Hi, testuser01 Logout

Welcome, testuser01

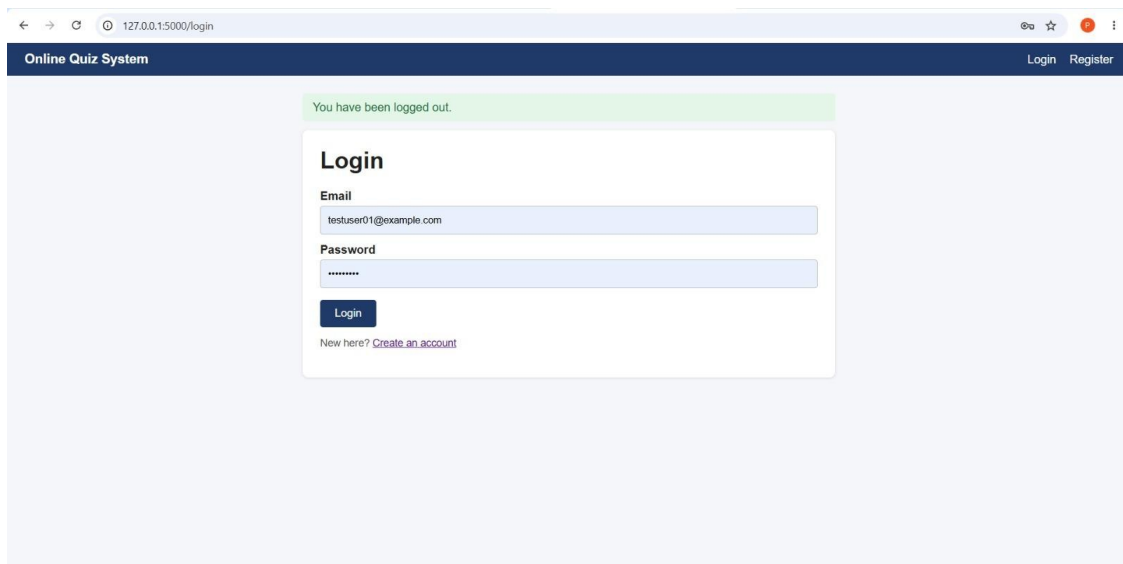
You are logged in as **User**.

Quiz categories will appear here (S1-03).

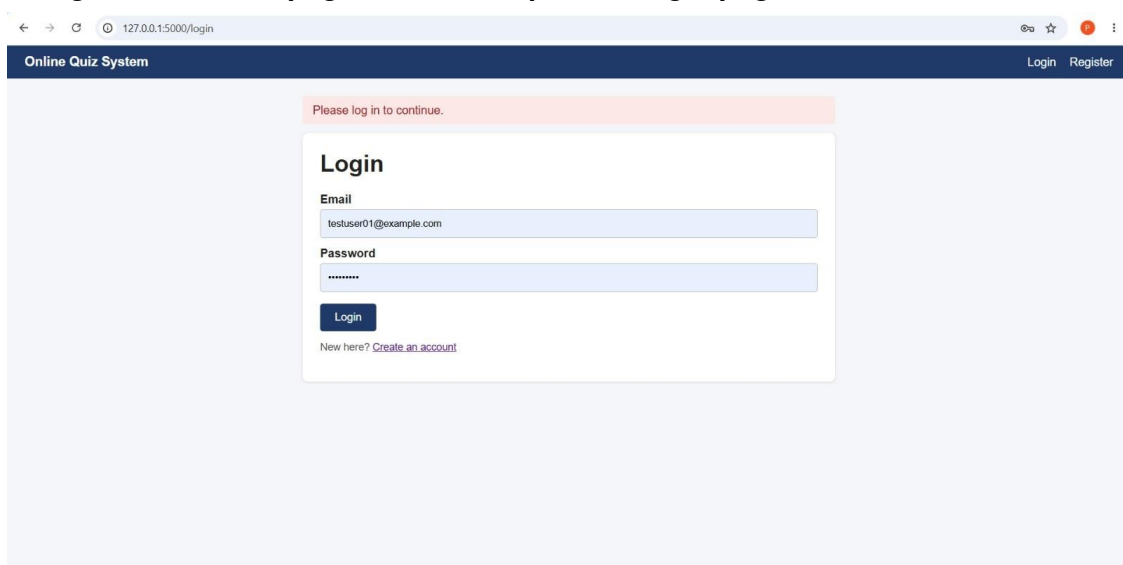
Logout successful. Login page is shown.



Logout message shown again.



After logout, the home page cannot be opened. Login page is shown.



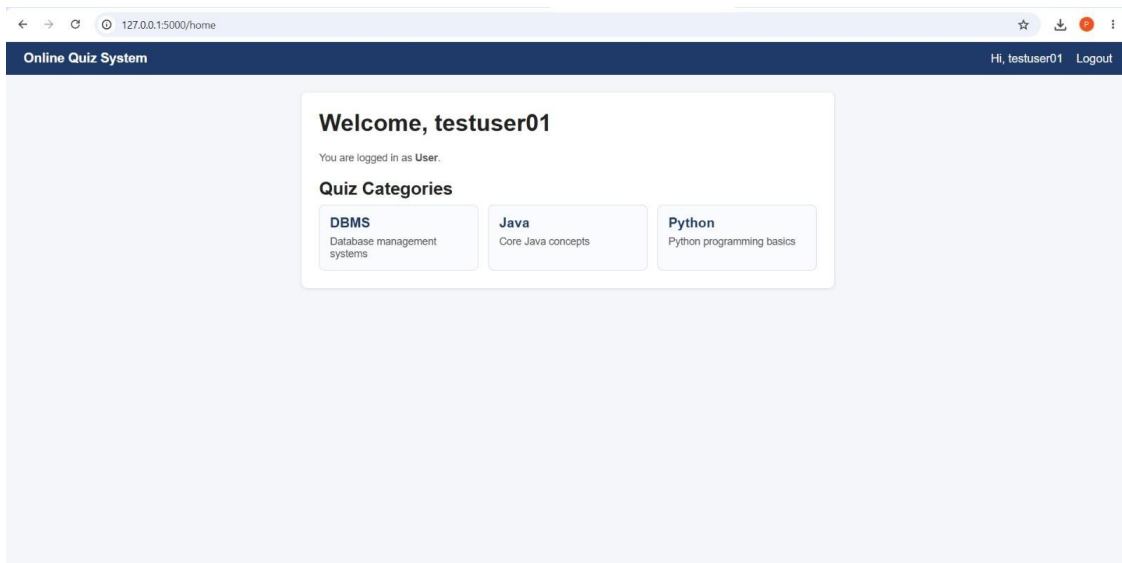
S1-03 Display Quiz Categories

Files unzipped and the server started.

```
(venv) PS C:\Users\DELL\Desktop\quiz_system> Expand-Archive "$HOME\Downloads\S1-03_categories_files.zip" -DestinationPath . -Force
(venv) PS C:\Users\DELL\Desktop\quiz_system> Select-String "SELECT category_id" app.py
app.py:124:      "SELECT category_id, name, description FROM Category ORDER BY name"

(venv) PS C:\Users\DELL\Desktop\quiz_system> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 296-466-228
```

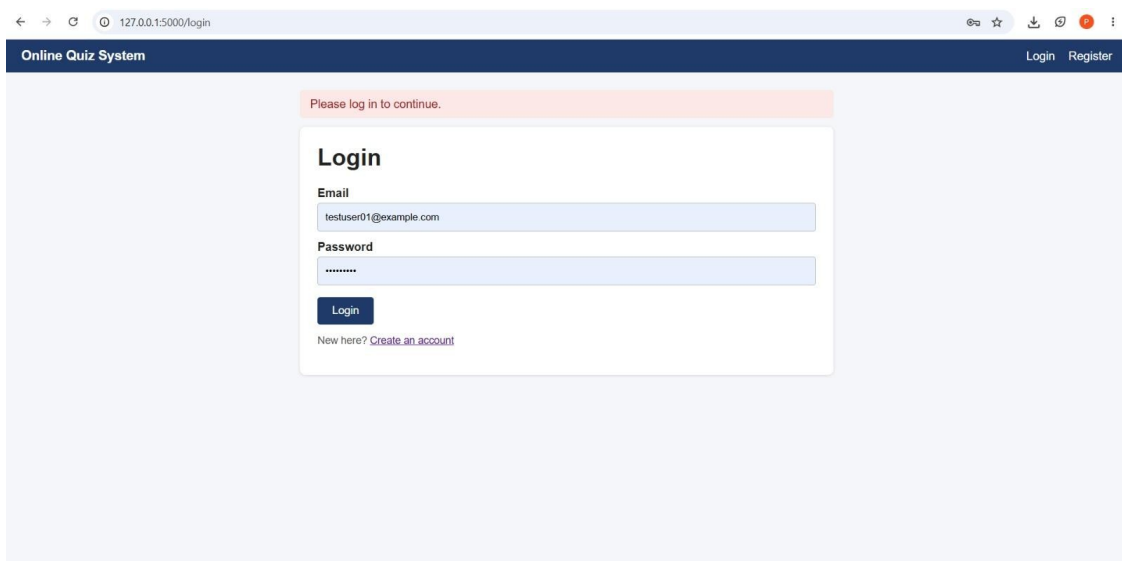
TC-01: Home page shows all categories.



Categories in the database match the home page.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r) for r in sqlite3.connect('quiz.db').execute('SELECT * FROM Category')]"
(1, 'Java', 'Core Java concepts')
(2, 'DBMS', 'Database management systems')
(3, 'Python', 'Python programming basics')
PS C:\Users\DELL\Desktop\quiz_system>
```

Home page needs login.



S1-04 Select a Quiz Category

Files unzipped and the server started.

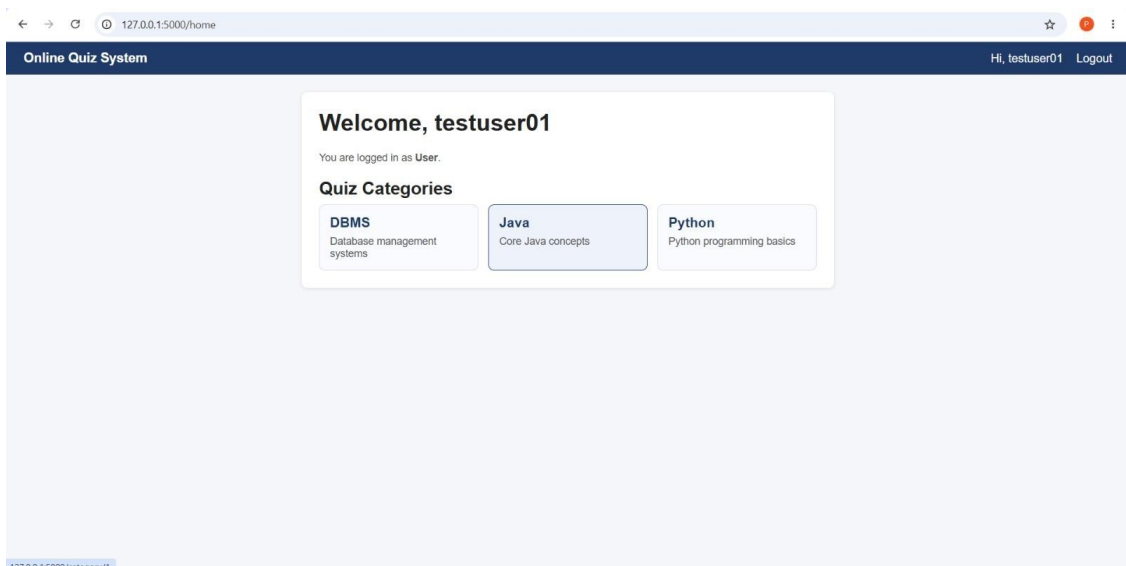
```
(venv) PS C:\Users\DELL\Desktop\quiz_system> Expand-Archive "$HOME\Downloads\S1-04_select_category_files.zip" -DestinationPath . -Force
(venv) PS C:\Users\DELL\Desktop\quiz_system> dir templates

Directory: C:\Users\DELL\Desktop\quiz_system\templates

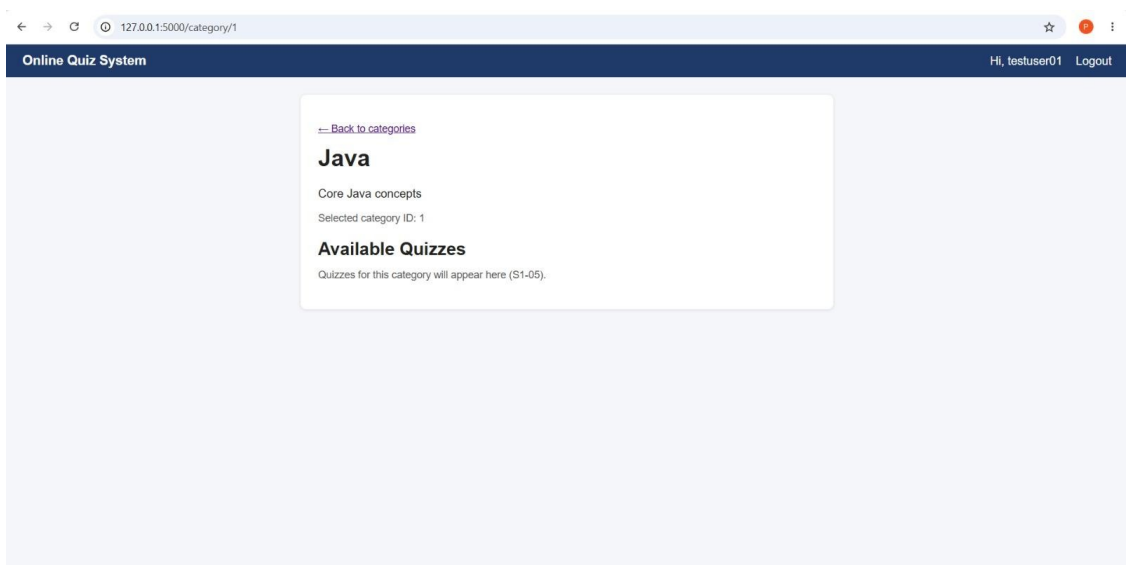
Mode                LastWriteTime         Length Name
----                -
-a-----         10/9/2026   6:15 AM             973 base.html
-a-----         10/9/2026   8:08 AM             519 category.html
-a-----         10/9/2026   8:08 AM             691 home.html
-a-----         10/9/2026   6:15 AM             552 login.html
-a-----         10/9/2026   6:15 AM             840 register.html

(venv) PS C:\Users\DELL\Desktop\quiz_system> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 296-466-228
```

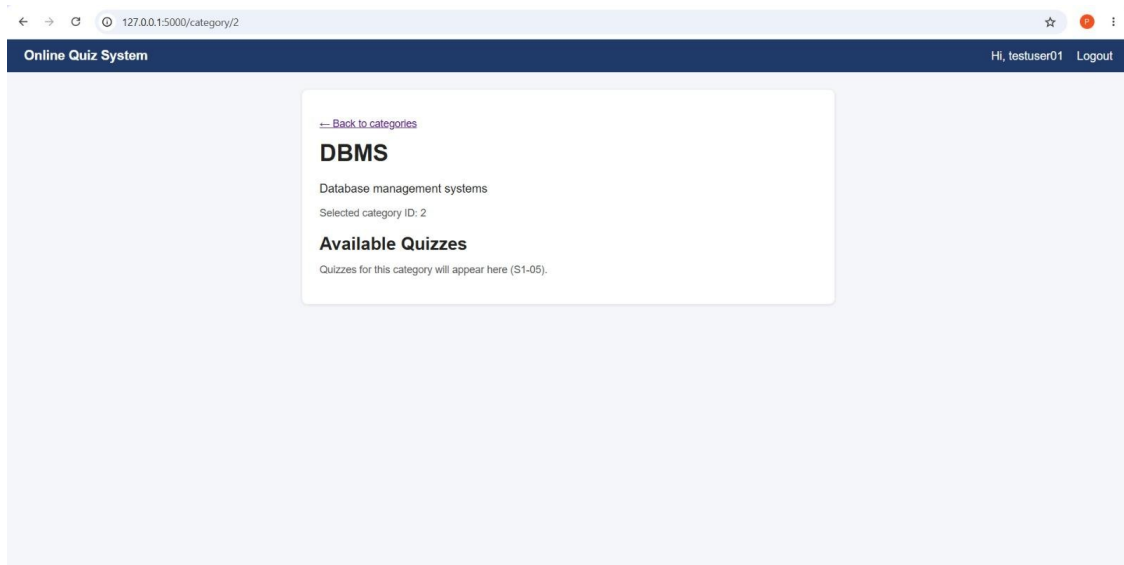
Category cards can be clicked.



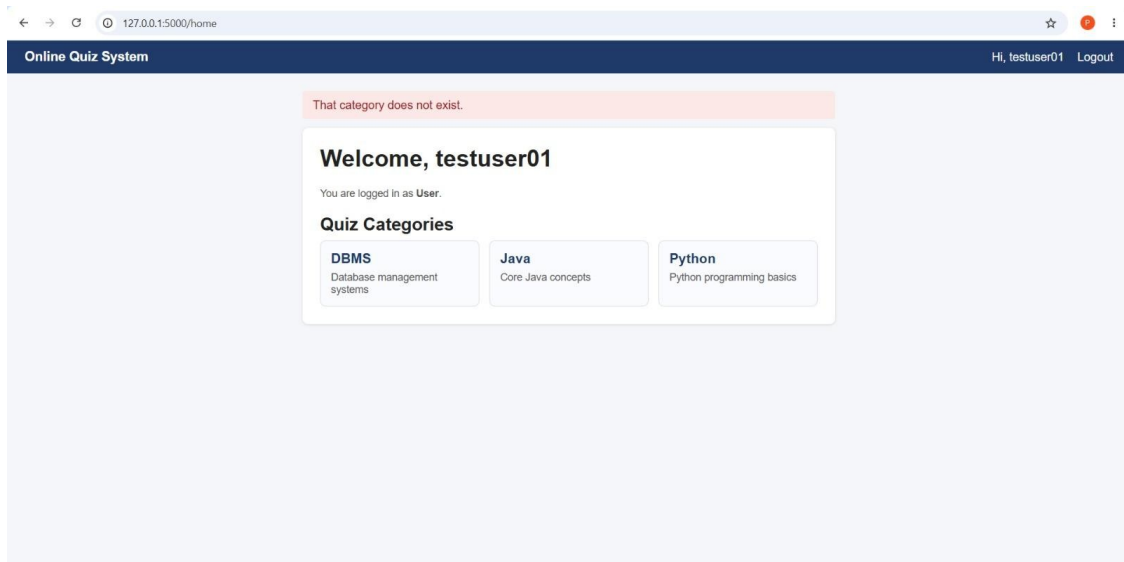
TC-02: Java category selected.



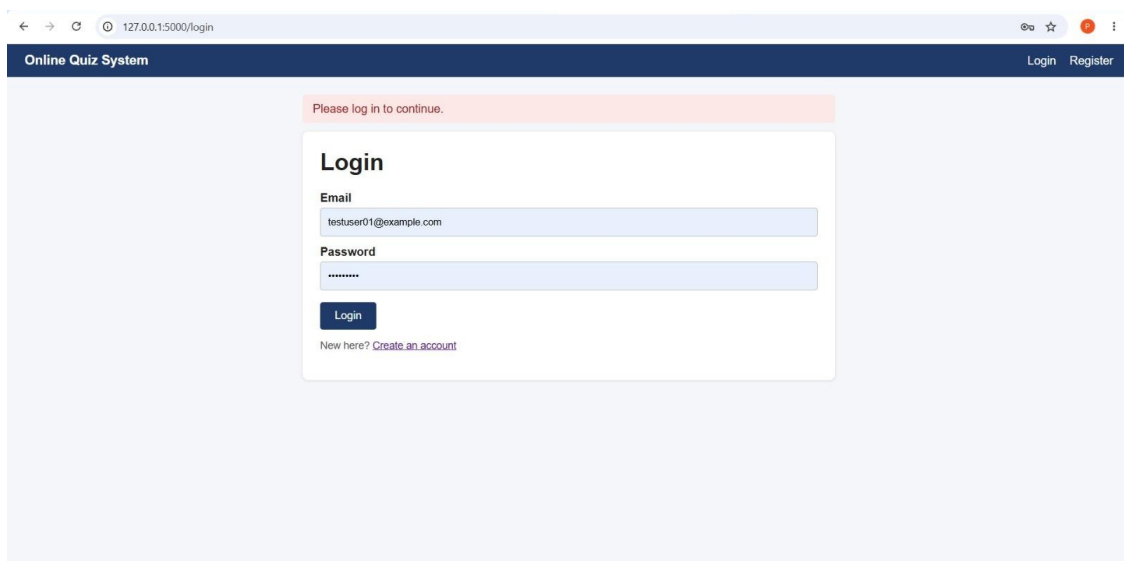
DBMS category selected.



Wrong category number shows an error message.



Category page needs login.



S1-05 Display Quizzes in a Category

Files unzipped and the server started.

```
(venv) PS C:\Users\DELL\Desktop\quiz_system> dir $HOME\Downloads\S1-05*

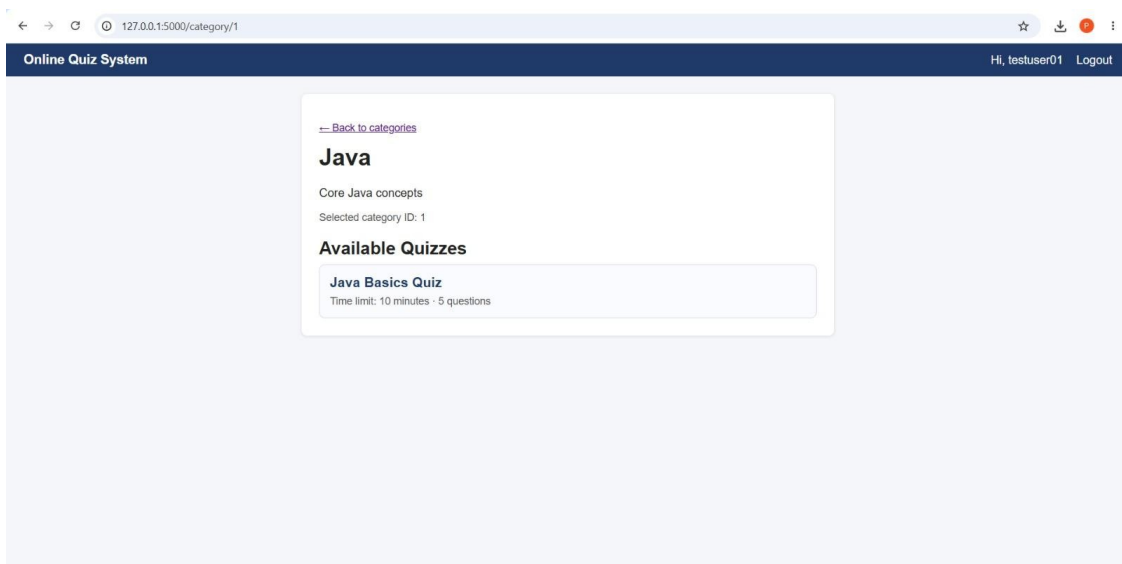
Directory: C:\Users\DELL\Downloads

Mode                LastWriteTime         Length Name
----                -
-a----          10/10/2026   8:35 AM           8598 S1-05_quizzes_files.zip

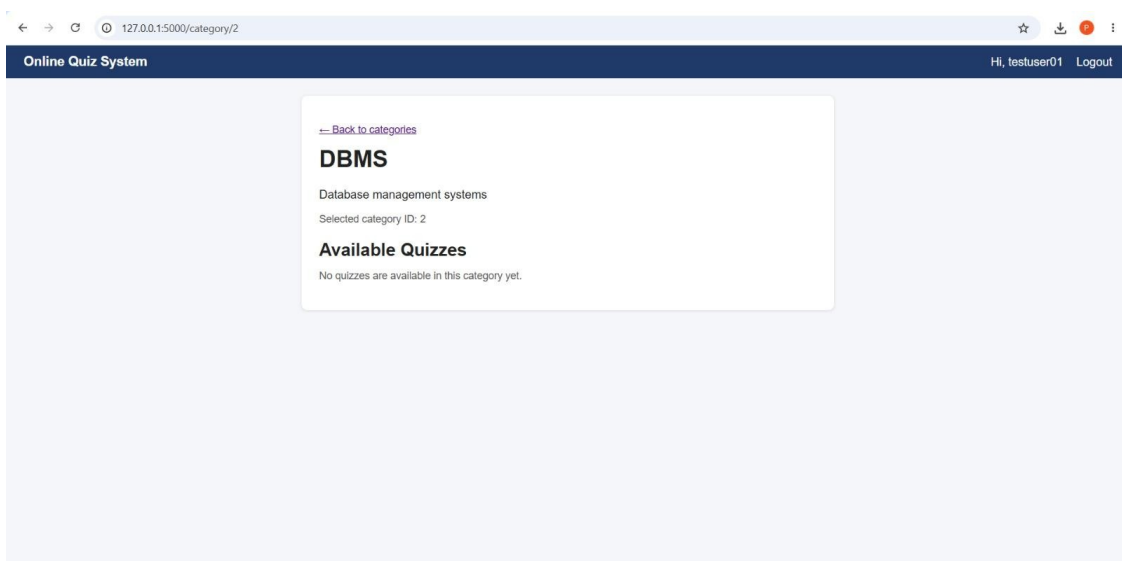
(venv) PS C:\Users\DELL\Desktop\quiz_system> Expand-Archive "$HOME\Downloads\S1-05_quizzes_files.zip" -DestinationPath .
-Force
(venv) PS C:\Users\DELL\Desktop\quiz_system> Select-String "FROM Quiz q" app.py
app.py:144:          FROM Quiz q

(venv) PS C:\Users\DELL\Desktop\quiz_system> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 296-466-228
```

TC-03: Java category shows its quiz.



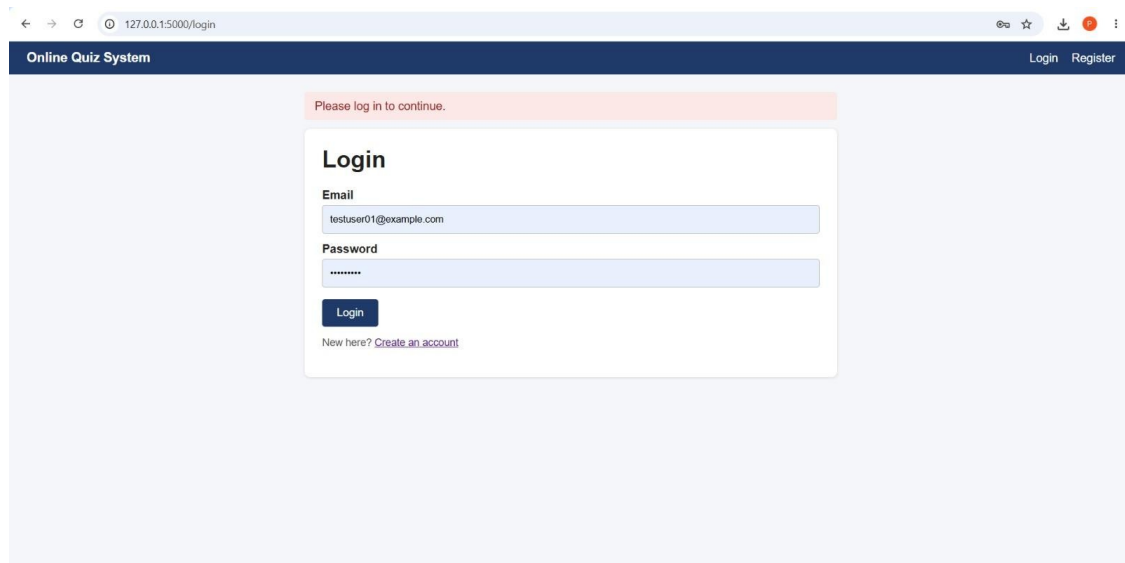
DBMS category has no quiz, so a message is shown.



Quiz in the database matches the Java page.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r) for r in sqlite3.connect('quiz.db').execute('SELECT quiz_id, title, time_limit, category_id FROM Quiz')]]"
(1, 'Java Basics Quiz', 10, 1)
PS C:\Users\DELL\Desktop\quiz_system> |
```

Category page needs login.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/login". The page title is "Online Quiz System". In the top right corner, there are links for "Login" and "Register". A red notification banner at the top of the main content area says "Please log in to continue." Below this, there is a white "Login" form. The form contains two input fields: "Email" with the value "testuser01@example.com" and "Password" with masked characters "*****". A blue "Login" button is positioned below the password field. At the bottom of the form, there is a link that says "New here? [Create an account](#)".

S1-06 Display Questions and Answer Options

Files unzipped and the server started.

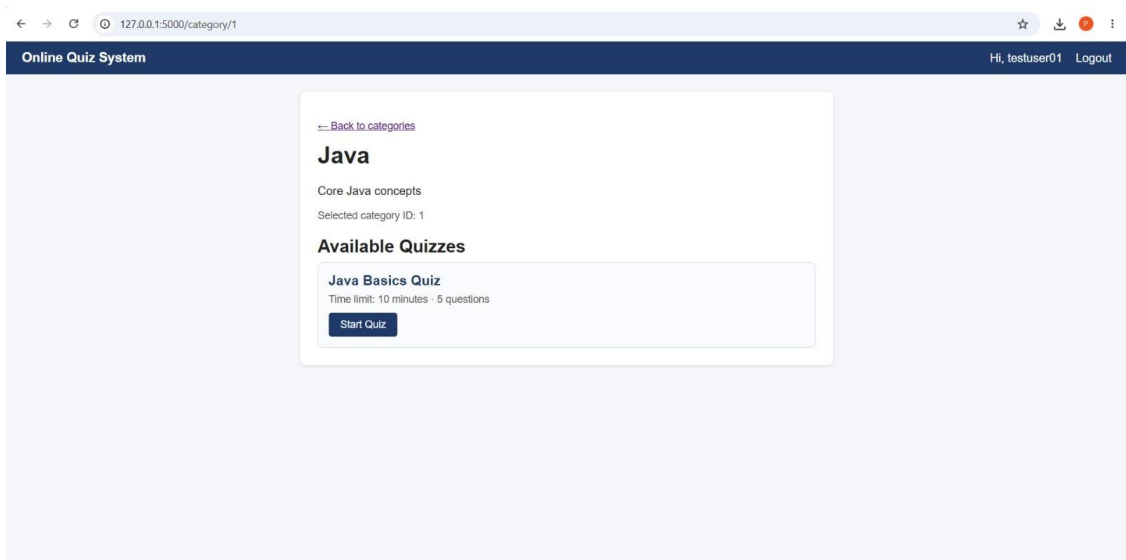
```
(venv) PS C:\Users\DELL\Desktop\quiz_system> Expand-Archive "$HOME\Downloads\S1-06_questions_files.zip" -DestinationPath . -Force
(venv) PS C:\Users\DELL\Desktop\quiz_system> dir templates

Directory: C:\Users\DELL\Desktop\quiz_system\templates

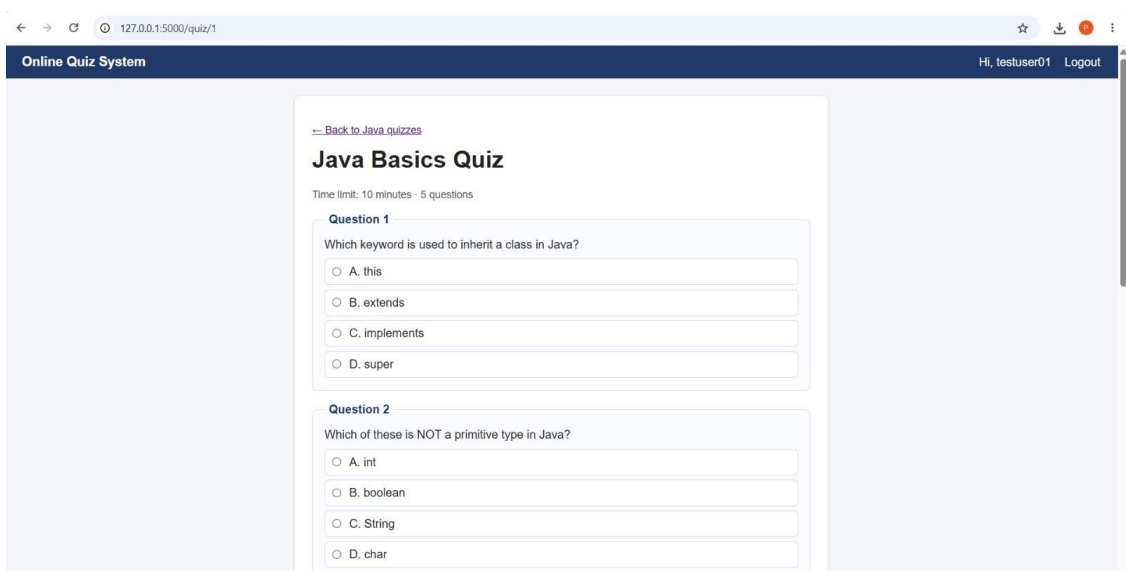
Mode                LastWriteTime         Length Name
----                -
-a-----         10/9/2026   6:15 AM           973 base.html
-a-----        10/10/2026   3:25 AM          1001 category.html
-a-----         10/9/2026   8:08 AM           691 home.html
-a-----         10/9/2026   6:15 AM           552 login.html
-a-----        10/10/2026   3:25 AM          1431 quiz.html
-a-----         10/9/2026   6:15 AM           840 register.html

(venv) PS C:\Users\DELL\Desktop\quiz_system> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 296-466-228
```

Java page shows the Start Quiz button.



TC-04: Quiz page shows the questions.



TC-05: Each question has four options. Submit button is shown.

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/quiz/1". The page content includes two questions, each with four radio button options. Question 4 asks "Which collection does not allow duplicate elements?" with options A. List, B. Set, C. ArrayList, and D. Vector. Question 5 asks "Which keyword prevents a class from being subclassed?" with options A. static, B. final, C. abstract, and D. private. A blue "Submit Quiz" button is located at the bottom of the question area.

One option can be selected for each question.

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/quiz/1". The page title is "Java Basics Quiz" with a subtitle "Time limit: 10 minutes - 5 questions". There is a link "Back to Java quizzes". The first question asks "Which keyword is used to inherit a class in Java?" with options A. this, B. extends (selected), C. implements, and D. super. The second question asks "Which of these is NOT a primitive type in Java?" with options A. int, B. boolean, C. String (selected), and D. char. The third question asks "What is the entry point method of a Java program?".

Questions in the database match the quiz page.

```
PS C:\Users\DELL\Desktop\quiz_system> py -c "import sqlite3; [print(r) for r in sqlite3.connect('quiz.db').execute('SELECT question_id, question_text FROM Question')]"
(1, 'Which keyword is used to inherit a class in Java?')
(2, 'Which of these is NOT a primitive type in Java?')
(3, 'What is the entry point method of a Java program?')
(4, 'Which collection does not allow duplicate elements?')
(5, 'Which keyword prevents a class from being subclassed?')
PS C:\Users\DELL\Desktop\quiz_system>
```

Quiz page needs login.

A screenshot of a web browser showing the login page of an 'Online Quiz System'. The browser's address bar displays '127.0.0.1:5000/login'. The page has a dark blue header with the site name and 'Login' and 'Register' links. A pink message box at the top says 'Please log in to continue.' Below this is a white login form with fields for 'Email' (containing 'testuser01@example.com') and 'Password' (masked with dots). A 'Login' button is at the bottom of the form, and a link for 'Create an account' is below it.

Online Quiz System Login Register

Please log in to continue.

Login

Email
testuser01@example.com

Password
.....

Login

New here? [Create an account](#)

Wrong quiz number shows an error message.

A screenshot of the 'Online Quiz System' home page. The browser's address bar shows '127.0.0.1:5000/home'. The header includes the site name and a user greeting 'Hi, testuser01' with a 'Logout' link. A pink message box at the top states 'That quiz does not exist.' Below this is a white welcome box for 'testuser01' that says 'You are logged in as User.' and lists 'Quiz Categories' in three boxes: 'DBMS' (Database management systems), 'Java' (Core Java concepts), and 'Python' (Python programming basics).

Online Quiz System Hi, testuser01 Logout

That quiz does not exist.

Welcome, testuser01

You are logged in as User.

Quiz Categories

DBMS
Database management systems

Java
Core Java concepts

Python
Python programming basics